

Using knit_expand() for templates

Yihui Xie

September 22, 2026

Contents

1	Write one row of data	1
1.1	Now i is 1	1
1.2	Now i is 2	1
1.3	Now i is 3	1
1.4	Now i is 4	2
1.5	Now i is 5	2
2	A regression model on several variables	2
2.1	Regression on cyl	2
2.2	Regression on disp	2
2.3	Regression on hp	3
2.4	Regression on drat	3
2.5	Regression on wt	3
2.6	Regression on qsec	4
2.7	Regression on vs	4
2.8	Regression on am	4
2.9	Regression on gear	5
2.10	Regression on carb	5
3	Multiple variables	5

1 Write one row of data

Only the first two sections are evaluated.

1.1 Now i is 1

This chunk is evaluated.

```
# row number 1
iris[1, ]
```

```
## Sepal.Length Sepal.Width Petal.Length Petal.Width Species
## 1           5.1           3.5           1.4           0.2 setosa
```

1.2 Now i is 2

This chunk is evaluated.

```
# row number 2
iris[2, ]

## Sepal.Length Sepal.Width Petal.Length Petal.Width Species
## 2           4.9           3           1.4           0.2 setosa
```

1.3 Now i is 3

This chunk is not evaluated.

```
# row number 3
iris[3, ]
```

1.4 Now i is 4

This chunk is not evaluated.

```
# row number 4
iris[4, ]
```

1.5 Now i is 5

This chunk is not evaluated.

```
# row number 5
iris[5, ]
```

2 A regression model on several variables

You can expand a template file, or just provide the template as a character string.

```
# the template
tpl = c("\\subsection{Regression on {{xvar}}}", "<<lm-{{xvar}}>>=",
        "lm(mpg~{{xvar}}, data=mtcars)", "@")
# expand to knitr source and pass to knit()
src = lapply(names(mtcars)[-1], function(xvar) {
  knit_expand(text = tpl)
})
```

2.1 Regression on cyl

```
lm(mpg ~ cyl, data = mtcars)

##
## Call:
## lm(formula = mpg ~ cyl, data = mtcars)
##
## Coefficients:
## (Intercept)          cyl
##      37.88      -2.88
```

2.2 Regression on disp

```
lm(mpg ~ disp, data = mtcars)

##
## Call:
## lm(formula = mpg ~ disp, data = mtcars)
##
## Coefficients:
## (Intercept)          disp
##    29.5999    -0.0412
```

2.3 Regression on hp

```
lm(mpg ~ hp, data = mtcars)

##
## Call:
## lm(formula = mpg ~ hp, data = mtcars)
##
```

```
## Coefficients:
## (Intercept)      hp
##      30.0989      -0.0682
```

2.4 Regression on drat

```
lm(mpg ~ drat, data = mtcars)

##
## Call:
## lm(formula = mpg ~ drat, data = mtcars)
##
## Coefficients:
## (Intercept)      drat
##      -7.52      7.68
```

2.5 Regression on wt

```
lm(mpg ~ wt, data = mtcars)

##
## Call:
## lm(formula = mpg ~ wt, data = mtcars)
##
## Coefficients:
## (Intercept)      wt
##      37.29      -5.34
```

2.6 Regression on qsec

```
lm(mpg ~ qsec, data = mtcars)

##
## Call:
## lm(formula = mpg ~ qsec, data = mtcars)
##
## Coefficients:
## (Intercept)      qsec
##      -5.11      1.41
```

2.7 Regression on vs

```
lm(mpg ~ vs, data = mtcars)

##
## Call:
## lm(formula = mpg ~ vs, data = mtcars)
##
## Coefficients:
## (Intercept)          vs
##      16.62         7.94
```

2.8 Regression on am

```
lm(mpg ~ am, data = mtcars)

##
## Call:
## lm(formula = mpg ~ am, data = mtcars)
##
## Coefficients:
## (Intercept)          am
##      17.15         7.24
```

2.9 Regression on gear

```
lm(mpg ~ gear, data = mtcars)

##
## Call:
## lm(formula = mpg ~ gear, data = mtcars)
##
## Coefficients:
## (Intercept)          gear
##       5.62         3.92
```

2.10 Regression on carb

```
lm(mpg ~ carb, data = mtcars)
```

```
##  
## Call:  
## lm(formula = mpg ~ carb, data = mtcars)  
##  
## Coefficients:  
## (Intercept)      carb  
##      25.87      -2.06
```

3 Multiple variables

```
tpl = "The value of a is {{a}} and b is {{b}}."
```

The value of a is 1 and b is 2.

The value of a is 5 and b is 2013.